Khalid Karroum

Data Science & Information Systems Security

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Available from January 2027 | Duration: 6 months

PROFESSIONAL PROFILE

Second-year Master's student in Data Science and Information Systems Security at Sultan Moulay Slimane University. Skilled in Python, SQL, exploratory data analysis, data preparation, and Machine Learning with scikit-learn and PyTorch. Developed projects in predictive modeling, anomaly detection, and data visualization, integrating REST APIs and Docker. Seeking a six-month final-year internship starting in January 2027 in Data Science, Machine Learning, or Data Analysis.

PROFESSIONAL EXPERIENCE

Data Intern | Completed internship

- Developed Price Monitor using Python and SQL to centralize international Brent crude oil, diesel, and bitumen prices and support purchasing analysis.
- Collected data from APIs and public sources, organized it in an SQL database, and tracked sources, dates, and units to compare quotations.
- Built a dashboard to monitor prices and historical trends, distinguishing market indices from supplier quotations to inform purchasing decisions.

TECHNICAL SKILLS

Programming & data: Python, SQL, pandas, NumPy, PostgreSQL, MySQL, MongoDB, ETL/ELT.

Machine Learning: scikit-learn, PyTorch, exploratory data analysis (EDA), data cleaning, feature engineering, classification, regression, clustering, cross-validation, calibration, model evaluation.

Applications & visualization: Django, FastAPI, REST APIs, Streamlit, Plotly.

MLOps & tools: Docker, Docker Compose, Git, GitHub Actions, CI/CD, automated testing, drift monitoring.

Security: Linux, Kali Linux, network security, cryptography.

PROJECTS

Aegis-Credit | Credit Risk and Model Governance

- Compared Logistic Regression, Random Forest, and Gradient Boosting; calibrated probabilities and evaluated models on an independent test set.
- Developed scoring through a REST API and CSV/XLSX batch processing using Django and PostgreSQL, with human review, drift monitoring, Docker Compose containerization, and continuous integration.

NetGuard | Network Anomaly Detection

- Applied unsupervised anomaly detection to UNSW-NB15 using Isolation Forest, LOF, and a PyTorch autoencoder; used labels only for evaluation.
- Implemented scoring via FastAPI, a Streamlit/Plotly dashboard, and Docker Compose containerization.

Telco Churn Dashboard | Customer Churn Prediction

- Built a scikit-learn pipeline with a stratified test split, 5-fold cross-validation, calibration, and threshold selection on training data; compared three classification models.
- Built Django and Streamlit interfaces sharing the same scoring logic; automated training and application tests with GitHub Actions.

EDUCATION

2025-2027 (expected) | Master's in Data Science & Information Systems Security (Excellence Program)
Sultan Moulay Slimane University, Beni Mellal, Morocco

2024-2025 | Bachelor's in Data Science & Information Systems Security (Excellence Program)
Sultan Moulay Slimane University, Beni Mellal, Morocco

2021-2024 | Diploma of General University Studies (DEUG) in Mathematics and Computer Science
Sultan Moulay Slimane University, Beni Mellal, Morocco

LANGUAGES

Arabic: Native | French: B2 | English: B2